The Representation Theory of Local Rules

Notes

Abstract

We explain the representation-theoretic content of Fomin's local rules for growth diagrams, at increasing levels of generality. At the most basic level, local rules encode the multiplicity-free branching rule for the symmetric groups. At the next level, they provide a bijective realisation of the Mackey formula for induction and restriction, which is equivalently the normal ordering relation in the Heisenberg algebra. The consistency of local rules across a growth diagram is the Yang–Baxter equation. In the cylindric setting, the higher-order local rules that transform bijections across adjacent profiles encode the action of the degenerate affine Hecke algebra, and the cylindric growth diagram is the combinatorial avatar of the Drinfeld functor to Yangian modules.

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1 Setup and notation

We work over a field k of characteristic zero. Partitions are denoted $\lambda, \mu, \nu, \ldots$ and written in weakly decreasing order $\lambda = (\lambda_1 \geqslant \lambda_2 \geqslant \cdots)$. We write $\lambda \vdash n$ for $|\lambda| = n$ and $\ell(\lambda)$ for the number of parts.

For a partition λ, define

$$U(\lambda) = \{\, \nu : \nu/\lambda \text{ is a horizontal strip} \,\} \quad \text{(partitions obtained by adding a horizontal strip)},$$
$$D(\lambda) = \{\, \mu : \lambda/\mu \text{ is a horizontal strip} \,\} \quad \text{(partitions obtained by removing a horizontal strip)}.$$

In the single-box case, $U(\lambda)$ consists of partitions obtained by adding one box and $D(\lambda)$ those obtained by removing one box.

We write S^λ for the Specht module (irreducible representation of S_n, where $\lambda \vdash n$) and W^λ for the Weyl module (irreducible polynomial representation of GL_N of highest weight λ).

2 Layer 1: Local rules as branching rules

2.1 The multiplicity-free branching rule

The restriction of an irreducible S_n-module to S_{n-1} is multiplicity-free:

$$S^\lambda\big|_{S_{n-1}} = \bigoplus_\mu S^\mu, \tag{1}$$

where μ ranges over all partitions obtained from λ by removing a single box. This multiplicity-freeness is the essential structural fact that makes local rules possible: at each step of a growth diagram, there is a *unique* box being added or removed, so the local rule can track it deterministically.

The same statement holds for the general linear groups: the branching $\mathrm{GL}_N \supset \mathrm{GL}_{N-1}$ gives

$$W^\lambda\big|_{\mathrm{GL}_{N-1}} = \bigoplus_\mu W^\mu,$$

where μ satisfies the interlacing condition $\lambda_1 \geqslant \mu_1 \geqslant \lambda_2 \geqslant \mu_2 \geqslant \cdots$. This is again multiplicity-free and underlies the Gel'fand–Tsetlin basis.

2.2 Fomin's local rule

A *growth diagram* is a rectangular grid whose vertices are labelled by partitions, with edges going up and to the right, such that adjacent partitions along any edge differ by at most one box. Each unit square has four corner partitions:

$$\begin{array}{ccc} \rho & \longrightarrow & \nu \\ \uparrow & x & \uparrow \\ \mu & \longrightarrow & \lambda \end{array}$$

Here $x \in \{0, 1\}$ is the entry of the input matrix at that cell. Fomin's *forward local rule* determines λ from ρ, μ, ν and x:

1. If $x = 1$ and $\mu = \nu = \rho$: then λ is ρ with one box added to the first row.
2. If $x = 0$ and $\mu = \nu = \rho$: then $\lambda = \rho$.
3. If $x = 0$ and $\mu = \rho \neq \nu$: then $\lambda = \nu$ (shadow passes vertically).
4. If $x = 0$ and $\nu = \rho \neq \mu$: then $\lambda = \mu$ (shadow passes horizontally).
5. If $x = 0$ and μ, ν, ρ pairwise distinct: then $\lambda = \mu \cup \nu$ (two shadows cross).
6. If $x = 0$, $\mu = \nu \neq \rho$, and μ differs from ρ in row r: then λ is μ with one box added to row $r + 1$ (two shadows bounce).

Proposition 2.1. *Fomin's local rule applied to a permutation matrix $\sigma \in S_n$ produces a growth diagram whose bottom-right boundary encodes the insertion tableau $P(\sigma)$ and whose top-right boundary encodes the recording tableau $Q(\sigma)$ of the Robinson–Schensted correspondence.*

2.3 Representation-theoretic content

Each case of the local rule corresponds to one step of the branching rule (1):

- **Cases 3–4** (shadow passes through): a restriction path from S^λ descends through S^μ on one side and S^ν on the other, and these paths are *compatible*—the same box is being tracked along both the S_n and GL_N chains.
- **Case 5** (crossing): two independent restriction paths pass through the cell without interacting. The box removed to get μ and the box removed to get ν are in different rows and columns, so they commute.
- **Case 6** (bouncing): two restriction paths arrive at the same partition $\mu = \nu$, having removed the same box from ρ. The local rule resolves this collision by adding the box back in the *next* available position (row $r + 1$). This is the non-trivial case—it encodes the fact that the branching multiplicity is exactly one.

The key point: the local rule exists *because* the branching is multiplicity-free. If $S^\lambda|_{S_{n-1}}$ had multiplicities, there would be no deterministic way to fill in the fourth corner.

3 Layer 2: Local rules as the Mackey formula

3.1 The general local rule

For semistandard tableaux and the Burge/RSK correspondences, the single-box local rule generalises to horizontal strips. A *forward local rule* is a bijection

$$\mathfrak{U}_{\alpha,\beta} \;:\; \big(\mathbb{Z}_{\geqslant 0}\,,\; D(\alpha) \cap D(\beta)\big) \;\xrightarrow{\sim}\; U(\alpha) \cap U(\beta) \tag{2}$$

satisfying the *weight condition*: if $\lambda = \mathfrak{U}_{\alpha,\beta}(m, \mu)$ then

$$|\lambda| + |\mu| = |\alpha| + |\beta| + m. \tag{3}$$

The *reverse local rule* is its inverse:

$$\mathfrak{D}_{\alpha,\beta} \;:\; U(\alpha)\cap U(\beta) \;\xrightarrow{\sim}\; \big(\mathbb{Z}_{\geqslant 0}\,,\; D(\alpha)\cap D(\beta)\big). \tag{4}$$

The integer m is the crucial new ingredient absent from the single-box case.

3.2 The Pieri operators

On the ring of symmetric functions Λ, define the *Pieri operator* $\Omega[Xv]$ and its dual $\Omega^*[Xu]$ by their action on the Schur basis:

$$\Omega[Xv]\cdot s_\mu = \sum_{\lambda\in U(\mu)} v^{|\lambda/\mu|}\, s_\lambda, \tag{5}$$

$$\Omega^*[Xu]\cdot s_\lambda = \sum_{\mu\in D(\lambda)} u^{|\lambda/\mu|}\, s_\mu. \tag{6}$$

These operators satisfy the commutation relation

$$\Omega^*[Xu]\;\Omega[Xv] \;=\; \frac{1}{1-uv}\;\Omega[Xv]\;\Omega^*[Xu]. \tag{7}$$

3.3 The local rule as bijective Mackey formula

Theorem 3.1. *The forward local rule $\mathfrak{U}_{\alpha,\beta}$ provides a bijective proof of the commutation relation* (7)*. Specifically, it witnesses the identity*

$$\langle s_\alpha \mid \Omega^*[Xu]\;\Omega[Xv]\; s_\beta\rangle \;=\; \frac{1}{1-uv}\,\langle s_\alpha \mid \Omega[Xv]\;\Omega^*[Xu]\; s_\beta\rangle$$

as a weighted bijection between the two sides, where the factor $\frac{1}{1-uv} = \sum_{m\geqslant 0}(uv)^m$ is accounted for by the integer m.

Now we interpret this representation-theoretically. The Pieri operator $\Omega[Xv]$ corresponds to **induction**: the Pieri rule $s_\mu\cdot h_k = \sum_\lambda s_\lambda$ (summing over horizontal strips λ/μ of size k) computes

$$\operatorname{Ind}_{S_n\times S_k}^{S_{n+k}}\big(S^\mu\boxtimes \mathbf{1}_{S_k}\big) \;=\; \bigoplus_{\lambda/\mu \text{ horiz. strip},\, |\lambda/\mu|=k} S^\lambda.$$

The dual operator $\Omega^*[Xu]$ corresponds to **restriction** followed by projection onto the trivial representation of the "removed" factor.

The commutation relation (7) therefore says:

"First induce, then restrict" equals $\frac{1}{1-uv}$ times "first restrict, then induce."

This is a **Mackey-type decomposition**. In the classical Mackey formula, the double cosets $S_{n-j}\backslash S_n/(S_{n-k}\times S_k)$ control the relationship between induction and restriction. The integer m in the local rule (2) is precisely the contribution from the "identity double coset"—the part where induction and restriction interact non-trivially.

Remark 3.2. The weight condition (3) has a natural interpretation: $|\lambda|-|\alpha|$ is the size of the horizontal strip added on the α-side, $|\beta|-|\mu|$ is the size of the strip removed on the β-side, and m is the deficit—the number of boxes that "pass through" from the removal to the addition without being accounted for by the strips. It is the Mackey correction term.

4 Layer 3: Local rules as normal ordering in the Heisenberg algebra

4.1 The Heisenberg algebra and Fock space

Consider the Heisenberg algebra $\mathfrak{h}$ generated by operators $a_n, a_n^\dagger$ ($n \geqslant 1$) satisfying

$$[a_m,\, a_n^\dagger] = m\,\delta_{m,n}, \qquad [a_m, a_n] = [a_m^\dagger, a_n^\dagger] = 0.$$

The *Fock space* $\mathcal{F}$ has a basis $\{|\lambda\rangle : \lambda \text{ a partition}\}$ and carries the bosonic representation of $\mathfrak{h}$ via

$$a_n^\dagger|\lambda\rangle = \sum_{\mu:\, \mu/\lambda \text{ is a border strip of size } n} (-1)^{\mathrm{ht}(\mu/\lambda)}\,|\mu\rangle,$$
$$a_n|\lambda\rangle = \sum_{\nu:\, \lambda/\nu \text{ is a border strip of size } n} (-1)^{\mathrm{ht}(\lambda/\nu)}\,|\nu\rangle.$$

Under the boson–fermion correspondence, the ring of symmetric functions Λ is identified with $\mathcal{F}$, with $s_\lambda \leftrightarrow |\lambda\rangle$.

4.2 Vertex operators and commutation

The Pieri operators of Section 3 are (exponential) vertex operators in this language. Writing

$$U(v) = \Omega[Xv] = \exp\Big(\sum_{n\geqslant 1} \frac{v^n}{n}\, a_n^\dagger\Big) \quad \text{(creation/induction)},$$
$$D(u) = \Omega^*[Xu] = \exp\Big(\sum_{n\geqslant 1} \frac{u^n}{n}\, a_n\Big) \quad \text{(annihilation/restriction)},$$

the commutation relation (7) becomes the standard normal ordering identity for vertex operators:

$$D(u)\,U(v) = \frac{1}{1-uv}\,U(v)\,D(u). \tag{8}$$

In the simplest case ($u = v = 1$, single-box operators), this reduces to Fomin's relation

$$DU - UD = \mathrm{Id},$$

which is the Heisenberg commutation relation in its most elementary form, and the algebraic reason that Young's lattice is a 1-differential poset.

4.3 Local rules as normal ordering

Theorem 4.1. *Each application of a local rule moves one creation operator past one annihilation operator, picking up the commutator $\frac{1}{1-uv}$. The full growth diagram computes the normal ordering of a product of vertex operators.*

Concretely: a growth diagram with m columns and n rows starts with a product $D(u_1)\cdots D(u_m)\,U(v_1)\cdots U(v_n)$ (all annihilation operators to the left). Filling in the grid

by local rules reorders this into the normal-ordered form $U(v_1)\cdots U(v_n)\,D(u_1)\cdots D(u_m)$ times a product of commutator factors $\prod_{i,j}\frac{1}{1-u_iv_j}$.

The partition at grid position (i,j) records the intermediate state after i annihilation operators have been moved past j creation operators. The integers $m_{i,j}$ produced by the local rules at each cell are the exponents in the commutator factors.

Remark 4.2. This explains why RSK "works": the Robinson–Schensted–Knuth correspondence is the combinatorial computation of normal ordering in the Heisenberg algebra. The insertion tableau P records the result of the creation operators (the GL_N side), the recording tableau Q records the result of the annihilation operators (the S_n side), and the growth diagram tracks the commutation step by step.

5 Layer 4: Growth diagram consistency is the Yang–Baxter equation

5.1 The consistency condition

A growth diagram is filled in one cell at a time, applying the local rule at each step. The fundamental property is:

Proposition 5.1 (Consistency)**.** *The result of filling a growth diagram is independent of the order in which cells are processed, provided each cell is processed only after its left and lower neighbours are known.*

This is a non-trivial statement. Consider a 2×1 strip of two horizontally adjacent cells:

$$\begin{array}{ccc} \alpha' & \longrightarrow & \beta' & \longrightarrow & \gamma' \\ \uparrow & & \uparrow & & \uparrow \\ \alpha & \longrightarrow & \beta & \longrightarrow & \gamma \end{array}$$

Processing left-then-right: first determine β' from (α',α,β), then determine γ' from (β',β,γ). This is unambiguous. But for a 2×2 block:

$$\begin{array}{ccccc} g & \longrightarrow & h & \longrightarrow & \mathbf{?} \\ \uparrow & & \uparrow & & \uparrow \\ d & \longrightarrow & e & \longrightarrow & f \\ \uparrow & & \uparrow & & \uparrow \\ a & \longrightarrow & b & \longrightarrow & c \end{array}$$

the cell marked **?** can be reached by two routes: first fill the top-left cell (e) then go right to get f, then up to get **?**; or first fill the bottom-right cell (e) then go up to get h, then right to get **?**. Consistency demands that both routes give the same answer.

5.2 The Yang–Baxter equation

This consistency condition is precisely the **Yang–Baxter equation** (YBE) in combinatorial form.

In Davydov–Molev's framework [2], the free symmetric monoidal category $\mathcal{S}$ is generated by one object X and an involutive Yang–Baxter operator $c : X \otimes X \to X \otimes X$ satisfying

$$(c \otimes 1_X)(1_X \otimes c)(c \otimes 1_X) = (1_X \otimes c)(c \otimes 1_X)(1_X \otimes c). \tag{9}$$

Under the symmetric fiber functor $F_N : \mathcal{S} \to \mathcal{V}ect$ (which sends $X \mapsto V = k^N$), the operator c becomes the swap map $\tau : v \otimes w \mapsto w \otimes v$ on $V^{\otimes 2}$.

Now consider the growth diagram as a lattice model. At each vertex, the local rule acts—it is the combinatorial R-matrix. The three strands in the YBE (9) correspond to three rows (or columns) of the growth diagram, and the two sides of the equation correspond to the two orders of processing a 2×2 block.

Theorem 5.2. *The local rule for growth diagrams satisfies the Yang–Baxter equation: for any valid boundary data on a 2×2 block, the partition at the final corner is independent of the order of evaluation.*

Remark 5.3. In the single-box case (Fomin's rules for permutations), the R-matrix is *involutive*: $c^2 = 1$. This corresponds to the classical case where the symmetric groups S_n are generated by involutions. In the Hecke algebra case, c satisfies $(c - q)(c + q^{-1}) = 0$ instead, giving a Yang–Baxter operator that is no longer involutive but satisfies a quadratic relation. The corresponding local rules are those of van Leeuwen's "Hecke insertion."

6 Layer 5: Cylindric local rules and the degenerate affine Hecke algebra

We now explain the deepest layer of the dictionary, connecting the higher-order local rules of [5] for cylindric partitions to the degenerate affine Hecke algebra Λ_n and the Drinfeld functor of Davydov–Molev [2].

6.1 The degenerate affine Hecke algebra

Definition 6.1. The *degenerate affine Hecke algebra* Λ_n is the associative algebra generated by elements $t_1, \ldots, t_{n-1}$ and $y_1, \ldots, y_n$ subject to:

$$t_i^2 = 1, \tag{10}$$
$$t_i t_{i+1} t_i = t_{i+1} t_i t_{i+1}, \tag{11}$$
$$t_i t_j = t_j t_i \quad \text{for } |i - j| > 1, \tag{12}$$
$$y_i t_i - t_i y_{i+1} = 1, \tag{13}$$
$$y_i y_j = y_j y_i, \tag{14}$$
$$y_j t_i = t_i y_j \quad \text{for } j \neq i, i{+}1. \tag{15}$$

As shown in [2], the degenerate affine Hecke algebras form a multiplicative sequence $\Lambda_* = \{\Lambda_n \mid n \geqslant 0\}$ via the maps $t_i \otimes 1 \mapsto t_i$, $1 \otimes t_j \mapsto t_{j+m}$, $y_i \otimes 1 \mapsto y_i$, $1 \otimes y_j \mapsto y_{j+m}$. The associated monoidal category $\mathcal{L} = \mathcal{C}(\Lambda_*)$ is the *degenerate affine Hecke category*, a free monoidal category generated by one object X, an endomorphism $x : X \to X$, and an involutive Yang–Baxter operator $c : X \otimes X \to X \otimes X$ satisfying (in addition to $c^2 = 1$ and the YBE):

$$(a \otimes b)\, c = c\, (b \otimes a), \qquad a, b \in A = k[y], \tag{16}$$

or equivalently $(x \otimes 1)\, c - c\,(1 \otimes x) = 1$.

6.2 Cylindric partitions and profiles

A *cylindric partition* of profile π is a sequence of partitions $(\mu^0, \mu^1, \ldots, \mu^T = \mu^0)$ such that consecutive pairs satisfy interlacing conditions determined by π. The profile π specifies, at each step i, whether μ^{i-1}/μ^i is a horizontal strip ($\pi_i = 0$, a "descent") or μ^i/μ^{i-1} is a horizontal strip ($\pi_i = 1$, an "ascent").

The generating function for cylindric partitions of profile π involves an alternating product of Pieri operators $\Omega[Xv_i]$ and dual Pieri operators $\Omega^*[Xu_j]$, applied cyclically around the cylinder.

6.3 The higher-order local rule

Definition 6.2 ([5])**.** The *higher-order local rule* $\mathfrak{L}_i$ transforms a bijection for cylindric partitions of profile π into a bijection for the adjacent profile π', where π' is obtained from π by transposing the i-th and $(i{+}1)$-th entries:

$$\mathfrak{L}_i : \big(\text{bijections for profile } \pi\big) \longrightarrow \big(\text{bijections for profile } \pi'\big).$$

Concretely, given a cylindric partition $\mathfrak{c} = (\mu^0, \ldots, \mu^T)$ of profile π', the rule $\mathfrak{L}_i$ applies the elementary local rule $\mathfrak{U}_{\mu^{i-1},\mu^{i+1}}$ to replace the i-th slice:

$$\mu^i \longmapsto \mathfrak{U}_{\mu^{i-1},\,\mu^{i+1}}(m, \mu^i) \quad \text{or} \quad (m, \mu^i) = \mathfrak{D}_{\mu^{i-1},\,\mu^{i+1}}(\mu^i),$$

depending on the direction of the profile change.

6.4 The representation-theoretic content

The dictionary between cylindric local rules and the degenerate affine Hecke algebra is:

Cylindric partitions	**Degenerate affine Hecke algebra**	**Representation theory**
Profile π	Ordering of y_i's and t_j's	Evaluation parameters
Adjacent transposition $\pi \to \pi'$	Generator $t_i \in \Lambda_n$	Swap of adjacent factors
Higher-order rule $\mathfrak{L}_i$	Action of t_i on modules	Elementary intertwiner
Evaluation parameter at slice i	Generator $y_i \in \Lambda_n$	Jucys–Murphy element
Consistency of $\mathfrak{L}_i\mathfrak{L}_{i+1}\mathfrak{L}_i$	Braid relation (11)	YBE
$\mathfrak{L}_i$ commutes with y_j $(j \neq i, i{+}1)$	Relation (15)	Locality
Profile rotation (cyclic shift)	Trace over auxiliary space	Cylindric character

Theorem 6.3 (Informal)**.** *The cylindric growth diagram, viewed as the composition of higher-order local rules $\mathfrak{L}_i$ that rotate the profile around the cylinder, is the combinatorial incarnation of the* Drinfeld functor

$$D_N : \mathcal{L} \to Y(\mathfrak{gl}_N)\text{-Mod}$$

from Davydov–Molev [2], evaluated on the vector representation $V \otimes k[y]$.

The argument is as follows:

1. The Drinfeld functor D_N sends the generator X of $\mathcal{L}$ to $V \otimes k[y]$, where $V = k^N$. The endomorphism $D_N(x)$ is multiplication by y in $k[y]$. The Yang–Baxter operator $D_N(c)$ is the involution $F_N(c) = \partial + t\tau$, where ∂ is the divided difference operator

$$\partial f(y_1, y_2) = \frac{f(y_1, y_2) - f(y_2, y_1)}{y_1 - y_2}$$

   and τ is the transposition of variables.

2. On tensor powers, $D_N(X^{\otimes n})$ is a module over the Yangian $Y(\mathfrak{gl}_N)$. The degenerate affine Hecke algebra Λ_n acts on $D_N(X^{\otimes n}) = (V \otimes k[y])^{\otimes n} \simeq V^{\otimes n} \otimes k[y_1, \ldots, y_n]$, with t_i acting by the divided difference operator on the i-th and $(i{+}1)$-th variables, and y_i acting by multiplication.

3. A cylindric partition of profile π with evaluation parameters $v_1, \ldots, v_n$ corresponds to a weight vector in this module. The slice sequence $(\mu^0, \mu^1, \ldots, \mu^T)$ records the branching path through the GL_N restriction chain, and the profile π specifies the ordering of creation and annihilation operators.

4. The higher-order local rule $\mathfrak{L}_i$ implements the action of t_i on this weight vector: it swaps two adjacent slices while applying the divided difference correction. The integer m produced by the elementary local rule is the divided difference contribution $\frac{f(y_i, y_{i+1}) - f(y_{i+1}, y_i)}{y_i - y_{i+1}}$.

5. The full rotation of the profile—composing $\mathfrak{L}_i$'s to cycle through all orderings—computes the trace of the product of R-matrices around the cylinder. This trace is the *transfer matrix* of the integrable lattice model, whose eigenvalues are the cylindric Schur functions.

6.5 Borodin's identity as Yangian integrability

The bijective proof of Borodin's identity in [5] works by iterating $\mathfrak{L}_i$ to transform a cylindric partition of one profile into one of another, accumulating integer partitions (the "inner partition" γ) along the way. The total weight is preserved at each step by the weight condition (3).

In the Yangian language, Borodin's identity

$$\sum_{\pi} \mathrm{CPP}(\pi;\, q) = \prod_{i,j} \frac{1}{1 - x_i y_j} \cdot \prod_{i<j} \frac{1}{(1 - x_i x_j)(1 - y_i y_j)}$$

is a statement about the **commutativity of transfer matrices**: the transfer matrices for different spectral parameters commute because the underlying R-matrix satisfies the Yang–Baxter equation. The product formula on the right-hand side arises from the eigenvalues of the commuting transfer matrices, computed via the Bethe ansatz or (in the degenerate case) the Yangian representation theory.

The bijective proof via local rules is the *combinatorial Bethe ansatz*: each application of $\mathfrak{L}_i$ is one step of diagonalising the transfer matrix, and the integers m produced at each step are the Bethe roots.

7 Summary: the five-layer dictionary

Combinatorics	**Representation theory**	**Algebra / Category theory**
Fomin local rule (single box)	Branching $S_n \downarrow S_{n-1}$	Restriction functor
General local rule $\mathfrak{U}_{\alpha,\beta}$	Mackey formula for Ind/Res	Commutation of induction and restriction functors
Pieri commutation	Normal ordering	Heisenberg algebra $[a, a^\dagger] = 1$
Growth diagram consistency	Yang–Baxter equation	R-matrix / Yang–Baxter object in $\mathcal{S}$
Higher-order rule $\mathfrak{L}_i$ (cylindric)	t_i action on Yangian modules	Drinfeld functor $D_N : \mathcal{L} \to Y(\mathfrak{gl}_N)$-Mod
Cylindric profile rotation	Transfer matrix trace	Commutativity of Yangian transfer matrices
Borodin's identity (bijective proof)	Integrability / Bethe ansatz	Categorical action of $\mathcal{L}$

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